

Conference Attendee Check-in Monitoring System

ASP.NET Core MVC Laboratory Exercise Specification

Objective

Develop an ASP.NET Core MVC application to monitor conference attendee check-in and check-out using Model Binding, Cookie Authentication, Validation, Repository Pattern, Bootstrap, CRUD, and Search.

Business Scenario

Event staff register attendees as they arrive, monitor attendees currently present, and record departure times.

Learning Objectives

MVC, Model Binding, DTOs, Validation, Cookie Authentication, Authorization, Repository Pattern, CRUD, Search, Bootstrap.

Functional Requirements

Register User, Login, Logout, Register Attendee, View Monitoring List, Edit Attendee, View Details, Record Check-Out, Search.

Attendee Fields

Ticket Number, First Name, Last Name, Company/School, Contact Number, Email, Event Name, Check-In Time, Check-Out Time, Notes.

Monitoring

Display Ticket Number, Attendee, Organization, Check-In, Check-Out, Status, Actions.

Status

Present, Left Event.

Validation

Use Required, EmailAddress, Phone, StringLength, Display, DataType and validate ModelState.

Authentication

Public: Login/Register. Protected: Conference pages using Cookie Authentication and [Authorize].

Entities

User: Id, FirstName, LastName, Email, Username, Password. AttendeeVisit: Id, TicketNumber, FirstName, LastName, Organization, ContactNumber, Email, EventName, CheckInTime, CheckOutTime, Status, Notes.

Repositories

UserRepository and AttendeeVisitRepository using static List.

Controllers & Views

Controllers: HomeController, AccountController, AttendeeController. Views: Login, Register, Index, Create, Edit, Details, CheckOut.

Submission

Complete MVC application with authentication, CRUD, validation, search, Bootstrap UI, repository pattern, and clean code.